Royal University of Phnom Penh
Faculty of Engineering

DATA SCIENCE AND ENGINEERING

Research topic:

Explainable AI for Automated Grading of Khmer Language Short-Answer Questions in Education

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Background and Motivation

- Short-answer questions are important because students explain their understanding in their own words, not only select an option.
- Manual grading takes time and can be inconsistent when teachers need to mark many answers.
- Automatic Short-Answer Grading has improved for high-resource languages, but Khmer still has limited NLP resources. Khmer is also difficult because it has no spaces between words and contains complex writing forms.



Research Statement

- There is no published Khmer ASAG benchmark.
- The dataset contains different maximum scores across questions, such as 5, 10, 15, and 20.
- Because the data comes from real classrooms, it also contains real grading noise, including cases where similar or identical answers receive different marks.

This research studies the comparation between four model families and explainable grading using real teacher-graded data.

Research Objective

- Collect and digitize handwritten real Khmer short-answer exam data
- Normalize scores so different question scales can be compared fairly
- Compare four model families: Classical, RNN, Transformer, and LLM
- Explain model predictions using SHAP word attribution
- Develop a prototype for score prediction and explanation

Research Questions

RQ1

Accuracy

Which model family gives the best agreement with the human teacher?

RQ2

Deployment

Which model is most useful for classroom deployment based on exact mark and within-one-point agreement?

RQ3

Explainability

Can SHAP provide understandable word-level explanations across all model families?

Literature Review

- ASAG research has developed from similarity-based methods to deep learning, Transformer models, and large language models.
- Most previous studies focus on high-resource languages, while Khmer short-answer grading is still missing.
- Some studies use explainable AI, but many systems still provide scores without clear word-level explanations.

Paper (author, year) and title	Language / dataset	Model & approach	XAI method	Headline result
Törnqvist et al. (2023) [14] *ExASAG: Explainable framework for automatic short answer grading*	English, short answer (ASAG)	Transformer grader + explanations	post-hoc XAI	per-score explanation for short-answer grading
Sung et al. (2022) [1] *A survey on automated short answer grading with deep learning: from word embeddings to transformers*	ASAG survey	Word embeddings → Transformers	n/a	survey (no single metric)
Chang & Ginter (2024) [10] *Performance of the large language model GPT-4 on automated short answer grading*	English, short answer	GPT-4 prompting (zero/few-shot)	n/a	QWK; fine-tuned specialists tend to lead
Soulimani et al. (2024) [18] *Deep learning based Arabic short answer grading in serious games*	Arabic, 3-class	Deep learning (Transformer)	n/a	Acc 95.67/77.22 (train/test); Cohen κ 0.60
Pinto & Shin (2025) [17] *Evaluating the consistency and reliability of attribution methods in ASAG systems*	English, ASAG	Attribution-method comparison	LOO, SHAP, IG	consistency / reliability of attributions
Xu et al. (2025) [16] *Explainable AI for education: enhancing essay scoring via rubric-aligned chain-of-thought prompting*	Essays (LLM)	Rubric-aligned chain-of-thought prompting	SHAP, LIME	rubric-aligned scoring; SHAP/ LIME weakly aligned
Lachana (2025) [15] *Automated scoring systems for open-ended questions in Dutch education*	Dutch, ordinal	Fine-tuned vs prompted LLM	n/a	QWK, exact/ adjacent; fine-tuned wins
Buoy et al. (2022) [3] *Pretrained models and evaluation data for the Khmer language*	Khmer NLP	Pretrained models + evaluation data	n/a	foundational Khmer models (no grading)

Methodology - Dataset Collection Process

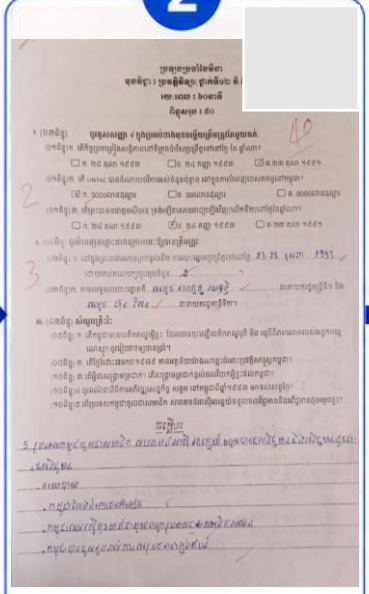
The dataset was built from real semester exam papers, not synthetic data.

1



Collect handwritten exam papers from two schools

2



Get real student answers and teacher marks

3



Manually digitize the Khmer handwritten answers into text

4

	A	B	C	D	E	F	G	H	I	J	K
1	SchoolID	ClassID	Subject	StudentID	QuestionID	Question	Reference	Answer	Student Score	Max Score	Year
2	AS	10C	Biology	S001	Q1	តើអាម៉ីបរស់នៅ	អាម៉ីបរស់		3	8	2026
3	AS	10C	Biology	S001	Q2	តើគោជែកប្រើសសុីថែ	គោជែកប្រ		7	8	2026
4	AS	10C	Biology	S001	Q3	តើល្អិតមានល្អិតមានរូបរ	ល្អិតមានរូ		8	10	2026
5	AS	10C	Biology	S001	Q4	ចូររៀបរាជ់ លក្ខណៈខុសគ្ន	ល្អិត-បន្តក		7	12	2026
6	AS	10C	Biology	S001	Q5	តើដំណាក់កាល ដំណាក់កា	ដំណាក់កា		8	15	2026
7	AS	10C	Biology	S002	Q1	តើអាម៉ីបរស់នៅ	អាម៉ីបរស់		8	8	2026
8	AS	10C	Biology	S002	Q2	តើគោជែកប្រើសសុីថែ	គោជែកប្រ		3	8	2026
9	AS	10C	Biology	S002	Q3	តើល្អិតមានល្អិតមានរូបរ	ល្អិតមានរូ		8	10	2026
10	AS	10C	Biology	S002	Q4	ចូររៀបរាជ់ លក្ខណៈខុសគ្ន	ល្អិត-បន្តក		5	12	2026
11	AS	10C	Biology	S002	Q5	តើដំណាក់កាល ដំណាក់កា	ដំណាក់កា		4	15	2026
12	AS	10C	Biology	S003	Q1	តើអាម៉ីបរស់នៅ	អាម៉ីបរស់		6	8	2026
13	AS	10C	Biology	S003	Q2	តើគោជែកប្រើសសុីថែ	គោជែកប្រ		8	8	2026
14	AS	10C	Biology	S003	Q3	តើល្អិតមានល្អិតមានរូបរ	ល្អិតមានរូ		8	10	2026
15	AS	10C	Biology	S003	Q4	ចូររៀបរាជ់ លក្ខណៈខុសគ្ន	លក្ខណៈខុ		10	12	2026
16	AS	10C	Biology	S003	Q5	តើដំណាក់កាល ដំណាក់កាល	រុក្ខជាតិមា		12	15	2026
17	AS	10C	Biology	S004	Q1	តើអាម៉ីបរស់នៅ	អាម៉ីបរស់		7	8	2026
18	AS	10C	Biology	S004	Q2	តើគោជែកប្រើសសុីថែ	គោជែកប្រ		6	8	2026
19	AS	10C	Biology	S004	Q3	តើល្អិតមានល្អិតមានរូបរ	ល្អិតមានរូ		4	10	2026
20	AS	10C	Biology	S004	Q4	ចូររៀបរាជ់ លក្ខណៈខុសគ្ន	លក្ខណៈខុ		11	12	2026
21	AS	10C	Biology	S004	Q5	តើដំណាក់កាល ដំណាក់កាល	រុក្ខជាតិមា		8	15	2026
22	AS	10C	Biology	S005	Q1	តើអាម៉ីបរស់នៅ	អាម៉ីបរស់		6	8	2026
23	AS	10C	Biology	S005	Q2	តើគោជែកប្រើសសុីថែ	គោជែកប្រ		5	8	2026
24	AS	10C	Biology	S005	Q3	តើល្អិតមានល្អិតមានរូបរ	ល្អិតមានរូ		4	10	2026
25	AS	10C	Biology	S005	Q4	ចូររៀបរាជ់ លក្ខណៈខុសគ្ន	លក្ខណៈខុ		12	12	2026
26	AS	10C	Biology	S005	Q5	តើដំណាក់កាល ដំណាក់កាល	រុក្ខជាតិមា		13	15	2026
27	AS	10C	Biology	S006	Q1	តើអាម៉ីបរស់នៅ	អាម៉ីបរស់		6	8	2026
28	AS	10C	Biology	S006	Q2	តើគោជែកប្រើសសុីថែ	គោជែកប្រ		5	8	2026

Store structured records: question, reference answer, student answer, teacher score, and maximum score

5



Clean and prepare the data for model training

Why Two Schools?

- We collect the data from two high schools: one from a **province** and one from the **capital city**.
- This is important because students from different locations may have different writing styles, answer quality, and classroom contexts.
- Using both province and city schools helps the dataset include more real educational variation.
- This makes the benchmark more realistic than collecting data from only one classroom or one school.



Dataset Overview

Final corpus summary:

- 1,184 answers
- 41 questions
- 203 students
- 2 schools
- 4 subjects: Biology, History, Geography, and Earth Science

Table 3: Khmer ASAG corpus statistics at a glance.

Property	Value	Notes
Answers	1,184	one incomplete row dropped
Questions	41	small question pool
Students	203	two schools (province + city)
Subjects	4	Biology 435, History 433, Geography 307, Earth Sci. 9
Max scores	$\{5, 6, 7, 8, 10, 12, 15, 20\}$	most common: 10 (404), 15 (306), 8 (135), 5 (130)
Grade labels (0–4)	14 / 153 / 327 / 191 / 499	skewed: $\approx$42 % receive full credit

Reference Answer and Score Normalization

- One challenge is that the teacher could not provide an official **reference answer** for every question.
- Therefore, for each question, the reference answer was selected from the **highest-scoring** student answer.
- However, the highest score does not always mean full marks. For example, the best available answer may be 8/10, not 10/10.
- After that, every teacher score is normalized:
- Normalized score = student score / maximum score
- Example: 2 / 10 = 0.20; label = round(4 × 0.20) = 1

Table 2: A real corpus record, shown verbatim (Biology; school AS, class 11K, student S056, question Q7). The student answer is vague and only partially correct, so the teacher awarded 2 of 5 points. English glosses are added for the reader and are not part of the data.

Field	Content
Question	ហេតុអ្វីបានជាមនុស្សស្រូបយកខ្យល់ដែលមានកាបូនម៉ូណូអុកស៊ីតអាចបង្កគ្រោះថ្នាក់ដល់ស្លាប់បាន? *(Why can breathing air that contains carbon monoxide be fatal to humans?)*
Reference	បានជាមនុស្សស្រូបយកខ្យល់ដែលមានកាបូនម៉ូណូអុកស៊ីដអាចបង្កឲ្យមានគ្រោះថ្នាក់ដល់ស្លាប់បានព្រោះកាបូនម៉ូណូអុកស៊ីដមានសមត្ថភាពចាប់យកក្នុងឈាមបានខ្លាំងជាងអុកស៊ីសែន វារារាំងឈាមមិនឲ្យដឹកនាំអុកស៊ីសែនទៅចិញ្ចឹមសរីរាង្គជាពិសេសខួរក្បាលដែលបណ្តាលឲ្យស្លាប់និងសន្លប់។ *(Because carbon monoxide binds in the blood more strongly than oxygen and blocks the blood from carrying oxygen to nourish the organs, especially the brain, causing unconsciousness and death.)*
Student answer	បានជាមនុស្សស្រូបយកខ្យល់ដែលមានកាបូនម៉ូណូអុកស៊ីដអាចបង្កគ្រោះថ្នាក់ដល់ស្លាប់បានព្រោះអុកស៊ីតដែលមនុស្សត្រូវការ ឧស្ម័នកាបូនិចនិងទឹក ទើបអាចរស់បាន។ *(Because the oxygen that humans need, carbonic gas and water, is what lets them live.)*
Score / Max	2 / 5 (normalised $y = 0.40$; grade label $\ell = \text{round}(4 \times 0.40) = 2$)

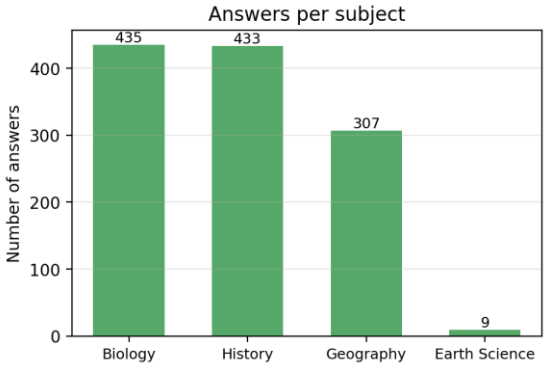
most answers are short, with a median near 193 characters and a long thin tail of a few very long answers.

Dataset Distribution and Label Noise

The dataset is imbalanced because Biology, History, and Geography have most records, while Earth Science has very few. Many answers also receive high scores.

The study also found real classroom issues:

- identical answers receiving different marks
- students rewriting the question as an answer
- 0-score cases when no meaningful content is provided



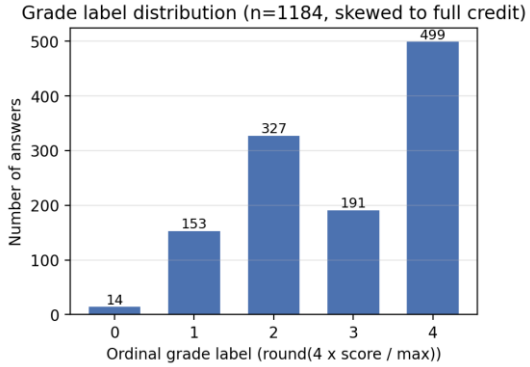


Table 6: Real records from the source spreadsheet (Class 10C Biology, Question 2; max score 8). Records 92 and 97 are identical yet scored 7 and 4; records 102 and 107 are identical yet scored 4 and 7. The same input mapped to different labels, so no model can learn a consistent rule from this subset.

Question 2 (max score 8): តើគេចែកប្រូទីសជាប៉ុន្មានក្រុម? អ្វីខ្លះ?

No.	Student answer	Score
92	គេចែកប្រូទីសសត្វជា ៤ប្រភេទគឺ-ប្រូទីសផ្លាស់ទីដោយជើងបញ្ឆោត : អាមីប-ប្រូទីស ផ្លាស់ទីដោយរោមញ័រ : ប៉ារ៉ាមេស៊ី-ប្រូទីសផ្លាស់ទីដោយ ប្លាសែល-ប្រូទីសផ្លាស់ទីដោយ ស្ប៉ូ។	7
97	គេចែកប្រូទីសសត្វជា ៤ប្រភេទគឺ-ប្រូទីសផ្លាស់ទីដោយជើងបញ្ឆោត : អាមីប-ប្រូទីស ផ្លាស់ទីដោយរោមញ័រ : ប៉ារ៉ាមេស៊ី-ប្រូទីសផ្លាស់ទីដោយ ប្លាសែល-ប្រូទីសផ្លាស់ទីដោយ ស្ប៉ូ។	4
102	គេចែកប្រូទីសសត្វជា ៤ ក្រុម គឺ -ណ្វៃយ៉ូ -ជើងបញ្ឆោតអាមីប -ស៊ីតូប្លាស -រ៉ាគុយអូលកន្ត្រាក់	4
107	គេចែកប្រូទីសសត្វជា ៤ ក្រុម គឺ -ណ្វៃយ៉ូ -ជើងបញ្ឆោតអាមីប -ស៊ីតូប្លាស -រ៉ាគុយអូលកន្ត្រាក់	7
177	គេចែកប្រូទីសសត្វជា ៣ ក្រុម -ប្រូទីសសត្វ -ប្រូទីសរុក្ខជាតិ - និងប្រូទីសផ្សិត។	7

F	G	H	I
Question	Answer	Student Scor	Max Scor
តើប្រព័ន្ធផលិតកម្មកសិកម្មនៅប្រទេសកម្ពុជាមានអ្វីខ្លះ?	ប្រព័ន្ធផលិតកម្មកសិកម្មនៅប្រទេសកម្ពុជាមាន	0	15
រៀបរាប់ពីទំនាក់ទំនងរវាងការដាំដុះអាជីវកម្មព្រៃឈើនិងនេសាទ។	ទំនាក់ទំនងរវាងការដាំដុះព្រៃឈើនិងការនេសាទ	0	15

Example from data: the answer repeats the question; teacher marked 0/15

Khmer-Aware Preprocessing

Khmer text is difficult because words are not separated by spaces. The text can also include invisible characters and punctuation issues.

Preprocessing includes:

- removing invisible characters
- removing punctuation
- optional Khmer word segmentation

Table 8: One answer (from Table 2) traced through the pipeline. `raw`/`clean` are shown in full and are deterministic; the `segment` tokens are produced by `khmer-nltk` (opening shown; the full segmentation is what the heatmaps render).

Stage	Result
`raw`	បានជាមនុស្សស្រូបយកខ្យល់ដែលមានកាបូនម៉ូណូអុកស៊ីដអាចបង្កគ្រោះថ្នាក់ដល់ស្លាប់បានព្រោះអុកស៊ីតដែលមនុស្សត្រូវការ ឧស្ម័នកាបូនិចនិងទឹក ទើបអាចរស់បាន។
`clean`	បានជាមនុស្សស្រូបយកខ្យល់ដែលមានកាបូនម៉ូណូអុកស៊ីដអាចបង្កគ្រោះថ្នាក់ដល់ស្លាប់បានព្រោះអុកស៊ីតដែលមនុស្សត្រូវការ ឧស្ម័នកាបូនិចនិងទឹក ទើបអាចរស់បាន (។ and invisibles removed)
`segment`	បាន ជា មនុស្ស ស្រូប យក ខ្យល់ ដែល មាន … (word boundaries inserted by `khmer-nltk`)

Overall Methodology Pipeline

- The full pipeline is:
- Dataset → Preprocessing → Train/Validation/Test Split → Model Training → Score Prediction → Evaluation → SHAP Explanation → Prototype
- Input: question, reference answer, student answer, teacher score, and maximum score.
- Output: predicted score, grade label, SHAP explanation, and feedback.

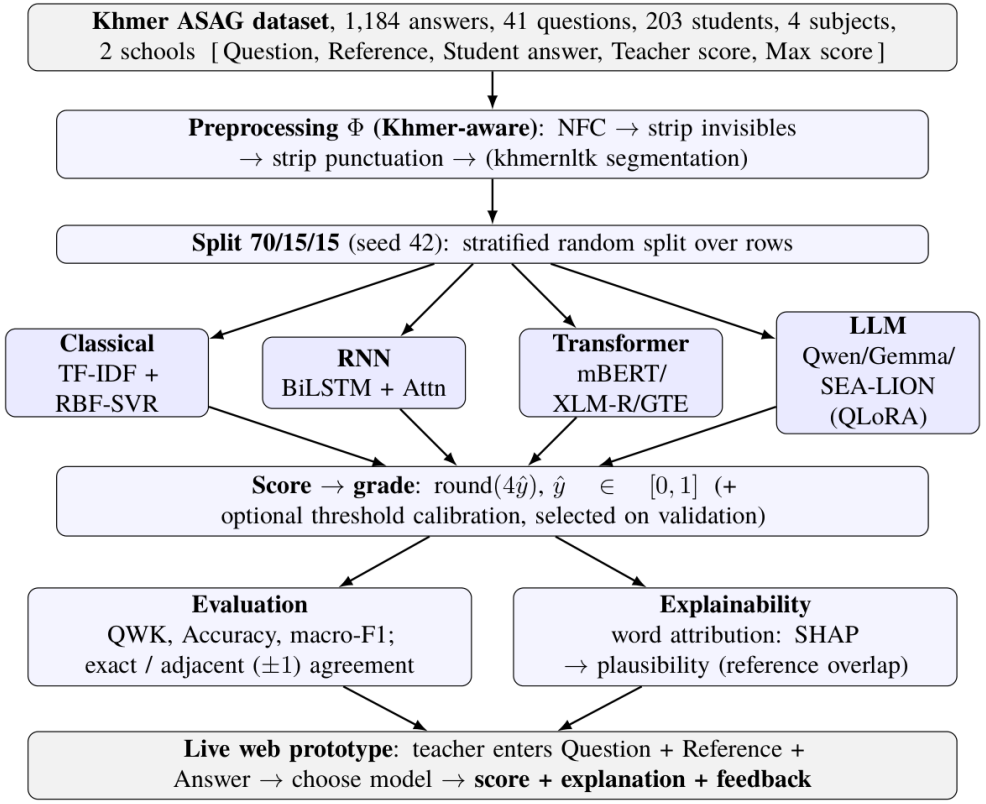


Figure 1: Overall research pipeline of this study.

Models Used

This study compares four model families:

1. Classical: TF-IDF with RBF-SVR
2. RNN: Character-level BiLSTM with Attention
3. Transformer: mBERT, XLM-R, and GTE
4. LLM: KhmerGrader using QLoRA fine-tuning

The purpose is to compare machine learning, deep learning, Transformer, and fine-tuned LLM approaches under one shared scoring task.

Table 15: Key hyperparameters of each model.

Pillar	Setting	Value
Classical	TF-IDF analyzer / n-grams	`char_wb`, $(2, 4)$
	max features / SVR C / kernel	15,000 / 1.0 / RBF
	extra baselines	`tfidf_cos`, `fasttext_cos` (100-dim)
RNN	char vocab / embed / hidden / layers	5,000 / 128 / 128 / 2 (bidirectional)
	dropout / batch / max epochs	0.3 / 64 / 20 (patience 4)
	learning rate / optimiser	1e$-$3 / Adam
Transformer	backbones	mBERT, XLM-R, GTE-multilingual
	config / frozen layers / max length / batch	dual + cross / 6 / 256 / 16
	learning rate / weight decay / dropout	2e$-$5 / 0.01 / 0.2
LLM	base / adaptation	Qwen 3.5 4B / QLoRA (4-bit NF4, double-quant)
	LoRA rank / alpha / dropout	16 / 16 / 0
	target modules	q, k, v, o, gate, up, down
	epochs / batch (effective) / warm-up	10 / 4 (16 via accumulation) / 5 %
	max sequence length / loss	1,024 / completion-only
	learning rate (val-selected)	$\in \{1, 2, 3\} \times 10^{-4}$

Evaluation Metrics

The study uses evaluation metrics such as:

QWK
Overall agreement with the teacher on ordinal grades.

Exact match
Same raw mark as the teacher.

Accuracy
Correct grade prediction.

Adjacent agreement
Within one point of the teacher.

Macro-F1
Balanced performance across all grade classes.

SHAP plausibility
Highlighted words overlap with reference content.

Experiment

We do the experiment using MISTI-HPC



CPU	32 CPU cores
RAM	64 GiB
GPU	NVIDIA A40
GPU memory	48 GB GDDR6 VRAM
Disk	536 GB

Main Results

KhmerGrader is the strongest overall model.

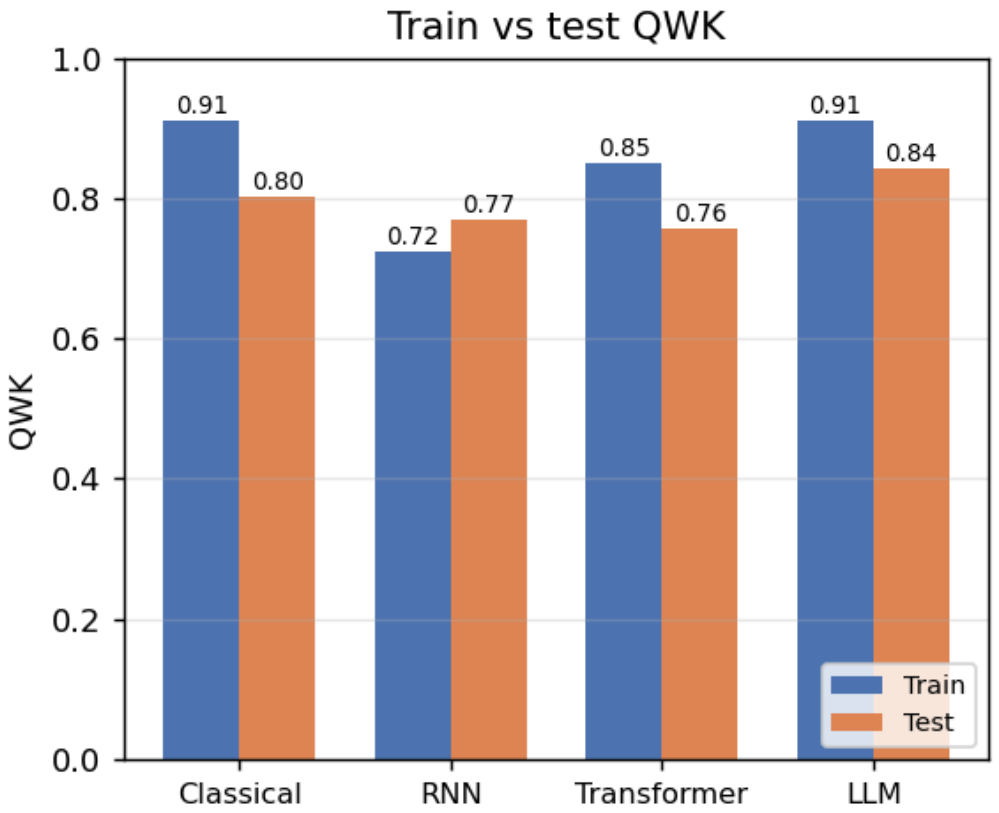
It achieves:

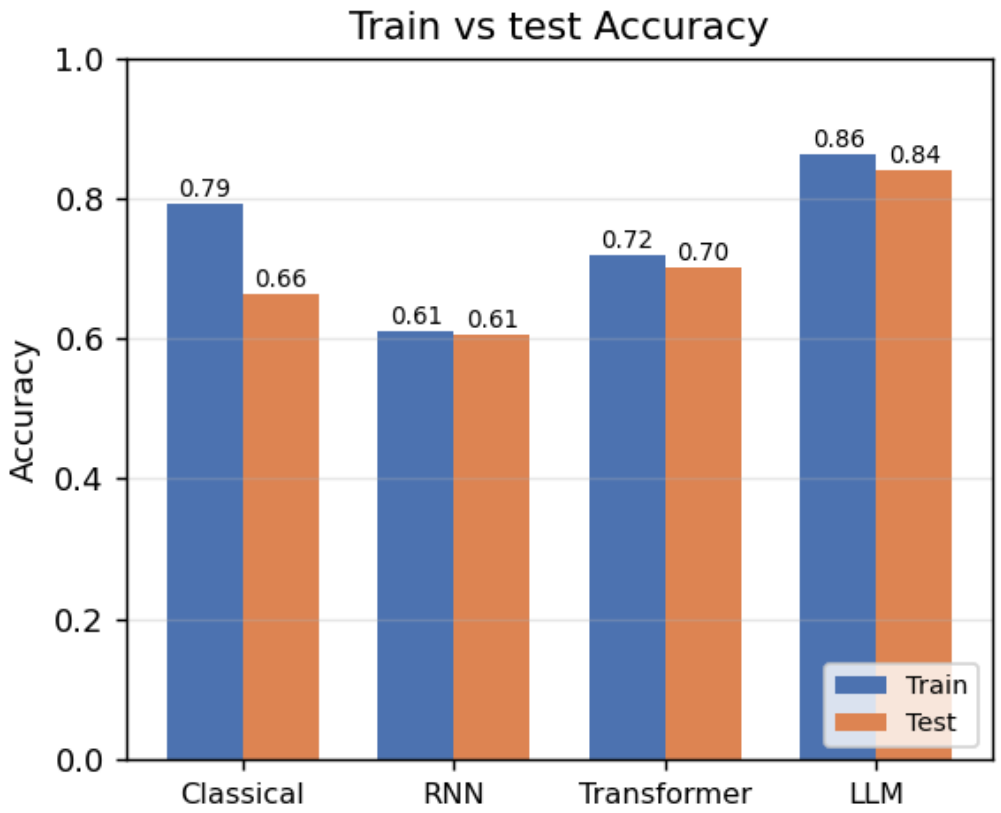
- QWK = 0.850
- Accuracy = 0.81
- Macro-F1 = 0.80
- Exact match = 0.67
- Adjacent agreement = 0.80

Table 10: Per-pillar champion results on the test set (uncalibrated, random split; macro-averaged F1). The fine-tuned LLM leads on every metric; the three smaller pillars cluster lower. The final row is the untrained Qwen base graded *zero-shot*, a reference that shows the fine-tuning lift, not a fifth pillar. QWK, accuracy, and macro-F1 are the champion-selection metrics; the exact integer-mark column is read from each champion's per-configuration leaderboard.

Pillar	QWK	Accuracy	macro-F1	Exact	Adjacent (± 1)
Classical (SVR)	0.802	0.65	0.47	0.28	0.72
RNN (BiLSTM)	0.789	0.69	0.53	0.34	0.70
Transformer (GTE)	0.777	0.68	0.51	0.49	0.66
LLM (Qwen-KhmerGrader, FT)	**0.850**	**0.81**	**0.80**	**0.67**	**0.80**
Qwen base, zero-shot (ref.)	0.529	0.42	0.30	0.25	0.43

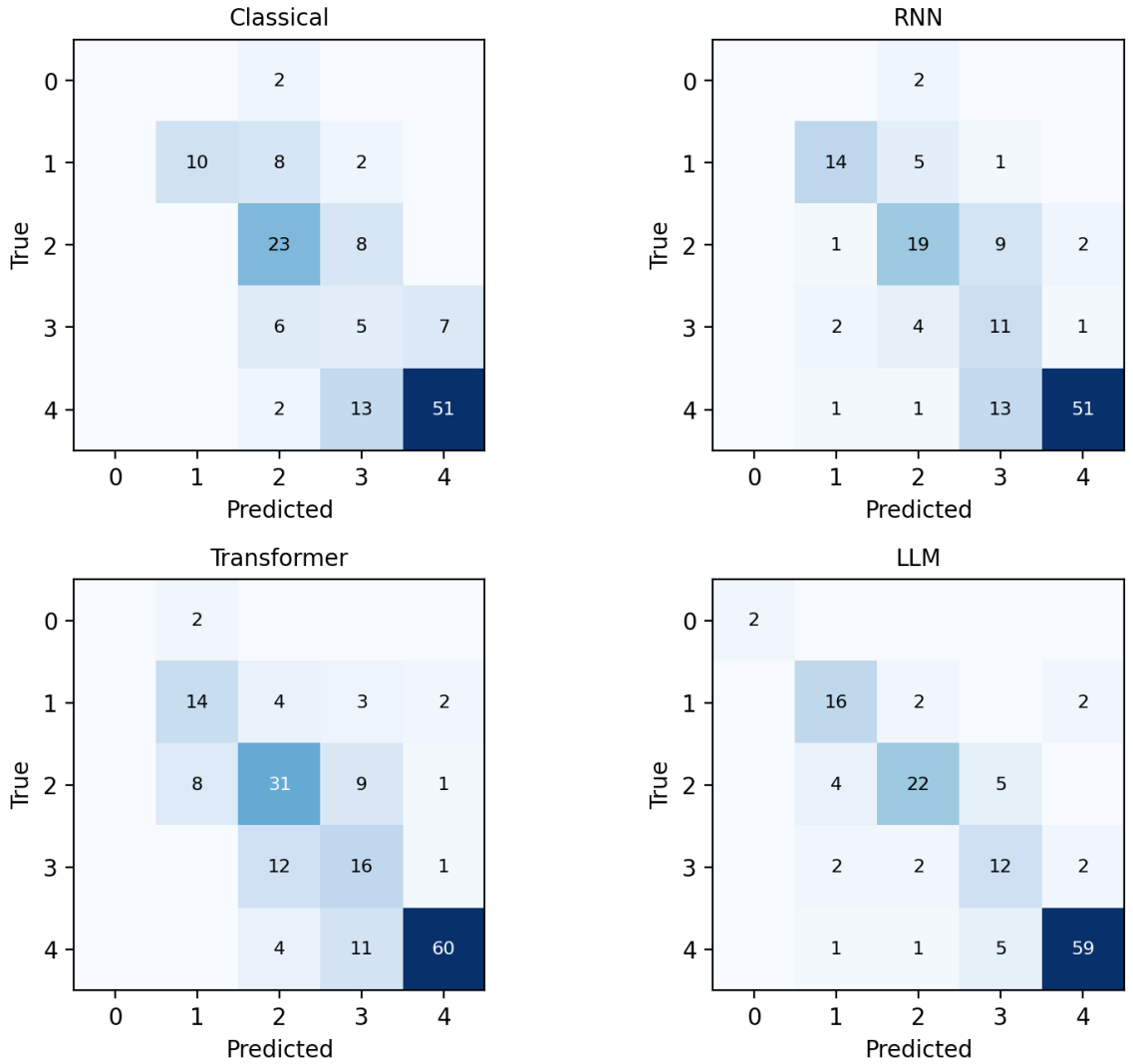
Results


Train vs test QWK
QWK
1.0
0.8
0.6
0.4
0.2
0.0
0.91
0.80
0.72
0.77
0.85
0.76
0.91
0.84
Train
Test
Classical
RNN
Transformer
LLM


Train vs test Accuracy
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
0.79
0.66
0.61
0.61
0.72
0.70
0.86
0.84
Train
Test
Classical
RNN
Transformer
LLM

Results

Confusion matrices (true vs predicted grade)



Fine-Tuning and Frontier Model Comparison

The original Qwen model performs lower in zero-shot grading. After QLoRA fine-tuning on real Khmer grading data, performance improves strongly.

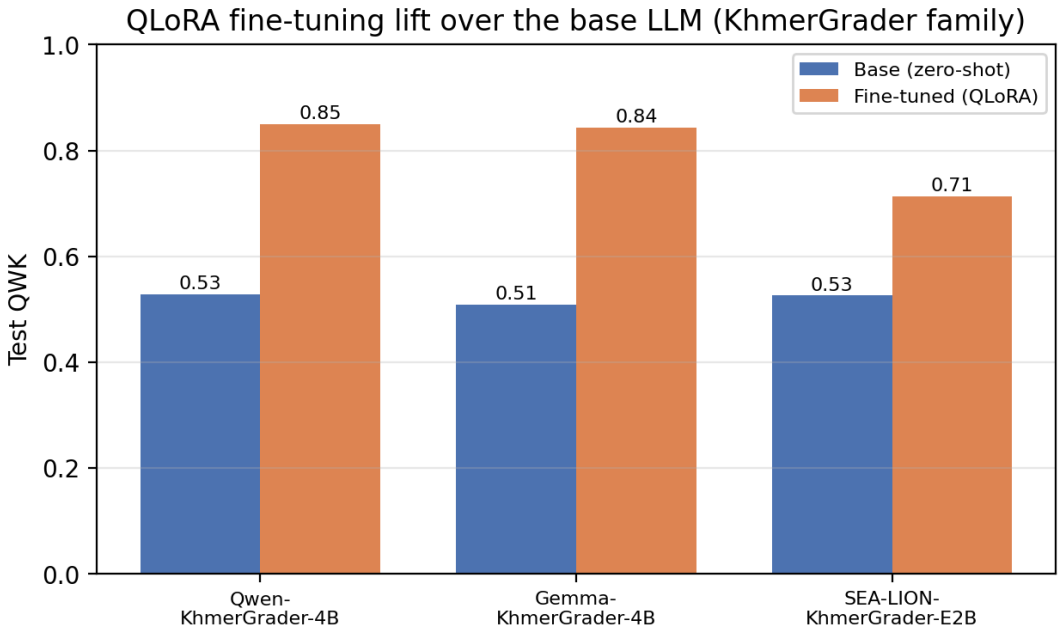


Table 12: KhmerGrader family: fine-tuned test QWK for each base (909 test set). The Qwen base scores 0.529 QWK *zero-shot* (Table 10), a +0.32 fine-tuning lift. The Qwen QWK is from `champion_metrics.csv`; the Gemma and SEA-LION QWK are those plotted in Figure 13 (from the exp08 fine-tune run).

KhmerGrader model	Base size	Fine-tuned QWK
Qwen-KhmerGrader-4B	4B	**0.850**
Gemma-KhmerGrader-4B	4B	0.844
SEA-LION-KhmerGrader-E2B	2.3B	0.713

Fine-Tuning and Frontier Model Comparison

KhmerGrader also outperforms larger frontier LLMs used with zero-shot prompting.

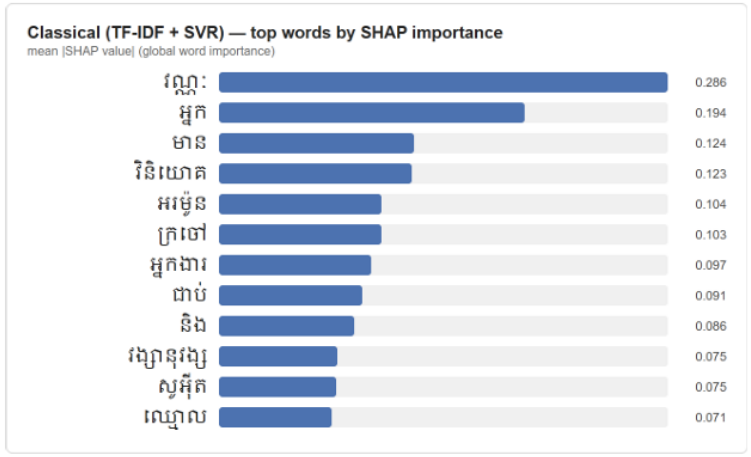
Table 13: Fine-tuned KhmerGrader champion vs. frontier LLMs prompted zero-shot on the `no10c` test set (137 answers; frontier metrics use each model's parsed subset, Appendix E), each in both the *bare* (integer answer) and *reasoning*-enabled settings. Frontier rows are a version-pinned API snapshot evaluated 2026-06-24; pinned versions: GPT `gpt-5.5-20260423`, Claude `claude-opus-4.8`, Gemini `gemini-3.5-flash-20260519`, DeepSeek `deepseek-v4-flash-20260423`. Metrics: QWK, accuracy (Acc), macro-F1, exact integer-mark, and within-one (± 1).

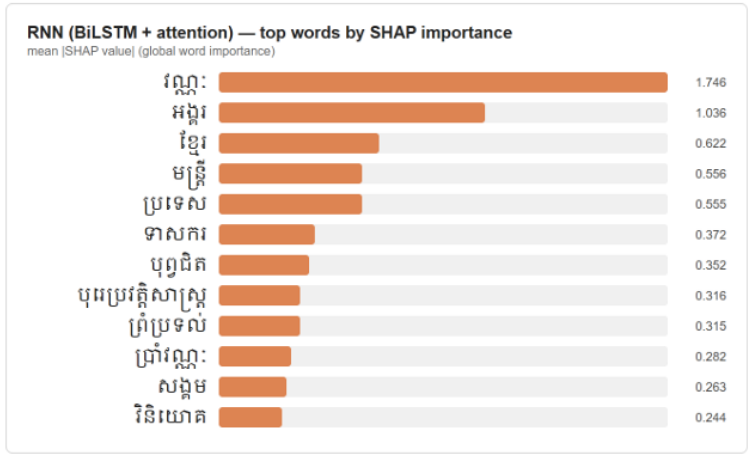
Model	Setting	QWK	Acc	F1	Exact	± 1
Qwen-KhmerGrader-4B	**fine-tuned**	**0.850**	**0.81**	**0.80**	**0.67**	**0.80**
GPT-5.5	bare	0.684	0.59	0.53	0.48	0.67
	reasoning	0.708	0.58	0.50	0.45	0.65
Claude Opus 4.8	bare	0.662	0.54	0.37	0.35	0.63
	reasoning	0.734	0.62	0.61	0.43	0.69
Gemini 3.5 Flash	bare	0.555	0.60	0.45	0.46	0.61
	reasoning	0.585	0.58	0.42	0.45	0.68
DeepSeek V4	bare	0.553	0.48	0.38	0.34	0.52
	reasoning	0.579	0.49	0.37	0.31	0.56

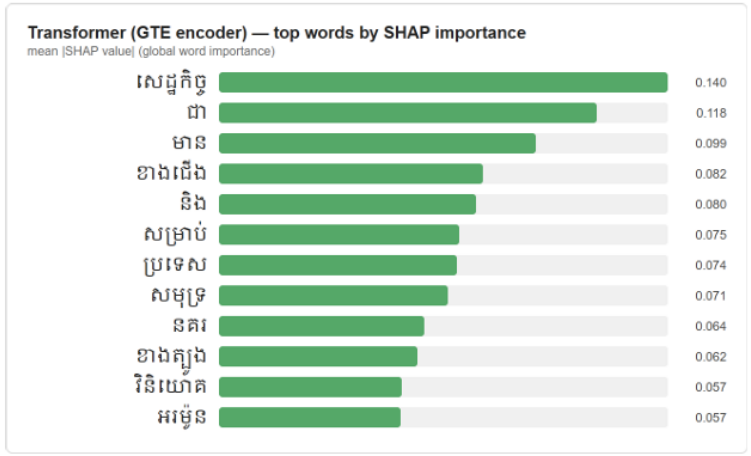
Key message: a smaller specialist model trained on Khmer data can outperform larger general models.

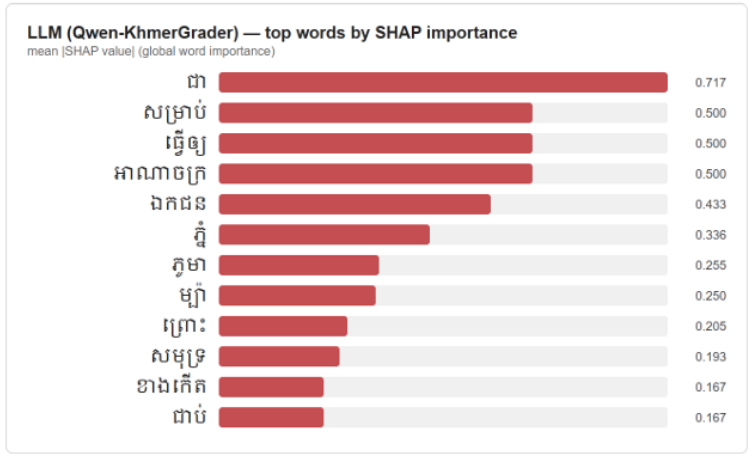
Explainability with SHAP

SHAP explains which words influence the predicted score. This helps teachers inspect whether the model focuses on content that matches the reference answer.

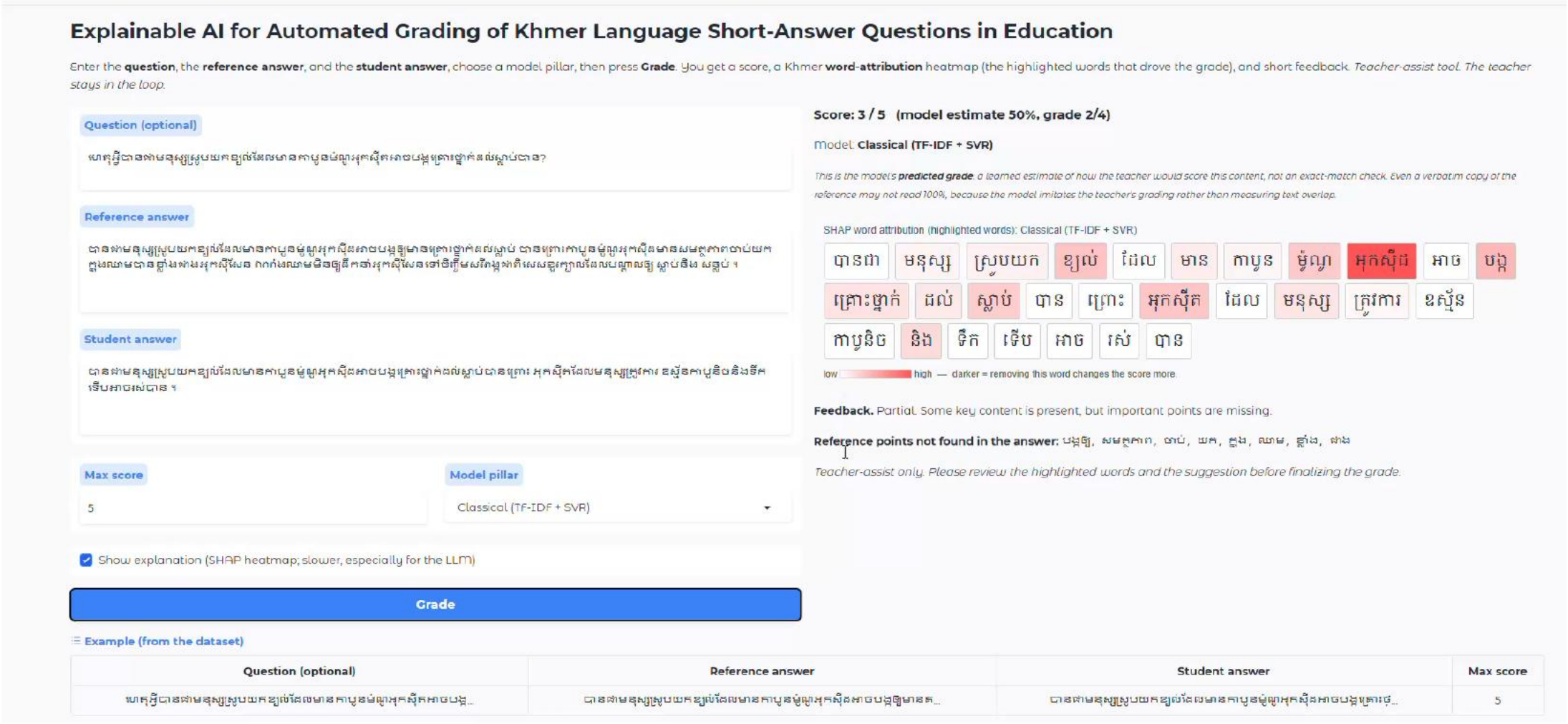








Prototype


Explainable AI for Automated Grading of Khmer Language Short-Answer Questions in Education
Enter the question, the reference answer, and the student answer, choose a model pillar, then press Grade. You get a score, a Khmer word-attribution heatmap (the highlighted words that drove the grade), and short feedback. Teacher-assist tool. The teacher stays in the loop.
Question (optional)
ហេតុអ្វីបានជាមនុស្សស្រូបយកខ្យល់ដែលមានកាបូនម៉ូណូអុកស៊ីតអាចបង្កគ្រោះថ្នាក់ដល់ស្លាប់បាន?
Reference answer
បានដោយមនុស្សស្រូបយកខ្យល់ដែលមានកាបូនម៉ូណូអុកស៊ីដអាចបង្កឱ្យមានគ្រោះថ្នាក់ដល់ស្លាប់ បានព្រោះកាបូនម៉ូណូអុកស៊ីដមានសមត្ថភាពជាប់យកក្តងឈាមបានខ្លាំងជាងអុកស៊ីសែន ពេលកំរិតឈាមមិនឲ្យដឹកជាអុកស៊ីសែនទៅចិញ្ចឹមសរីរាង្គនានាសេសខ្លះក្បាលដែលបណ្តាលឲ្យ ស្លាប់ធំង សន្លប់ ។
Student answer
បានដោយមនុស្សស្រូបយកខ្យល់ដែលមានកាបូនម៉ូណូអុកស៊ីដអាចបង្កគ្រោះថ្នាក់ដល់ស្លាប់បានព្រោះ អុកស៊ីតដែលមនុស្សត្រូវការ ខន្ទុកាបូនិចនិងទឹក ទើបអាចរស់បាន ។
Max score
5
Model pillar
Classical (TF-IDF + SVR)
Show explanation (SHAP heatmap; slower, especially for the LLM)
Grade
Score: 3 / 5 (model estimate 50%, grade 2/4)
Model: Classical (TF-IDF + SVR)
This is the model's predicted grade: a learned estimate of how the teacher would score this content, not an exact-match check. Even a verbatim copy of the reference may not read 100%, because the model imitates the teacher's grading rather than measuring text overlap.
SHAP word attribution (highlighted words): Classical (TF-IDF + SVR)
បានជា មនុស្ស ស្រូបយក ខ្យល់ ដែល មាន កាបូន ម៉ូណូ អុកស៊ីដ អាច បង្ក
គ្រោះថ្នាក់ ដល់ ស្លាប់ បាន ព្រោះ អុកស៊ីត ដែល មនុស្ស ត្រូវការ ខន្ទុ
កាបូនិច និង ទឹក ទើប អាច រស់ បាន
low high — darker = removing this word changes the score more.
Feedback. Partial. Some key content is present, but important points are missing.
Reference points not found in the answer: បង្កឱ្យ, សមត្ថភាព, ចាប់, យក, ក្តង, ឈាម, ខ្លាំង, ជាង
Teacher-assist only. Please review the highlighted words and the suggestion before finalizing the grade.
Example (from the dataset)
Question (optional)
Reference answer
Student answer
Max score
ហេតុអ្វីបានជាមនុស្សស្រូបយកខ្យល់ដែលមានកាបូនម៉ូណូអុកស៊ីដអាចបង្ក...
បានដោយមនុស្សស្រូបយកខ្យល់ដែលមានកាបូនម៉ូណូអុកស៊ីដអាចបង្កឱ្យមានគ...
បានដោយមនុស្សស្រូបយកខ្យល់ដែលមានកាបូនម៉ូណូអុកស៊ីដអាចបង្កគ្រោះថ្...
5

Limitations

- Small dataset from four subjects only
- Single-teacher grading
- Some real label noise
- Official reference answers were not available
- SHAP explanations and the prototype still need validation with real teachers.

This is a teacher-assist tool not autonomous grader
“Must have HUMAN-IN-THE-LOOP grading support”

Future Work

- Add more graders and larger datasets
- Improve Khmer-specific models
- Validate SHAP and prototype with teachers
- Improve student feedback quality

Conclusion

- This study built a reproducible Khmer ASAG benchmark using real teacher-graded answers.
- It compared four model families under one shared pipeline.
- KhmerGrader achieved the best accuracy and classroom deployment performance.
- SHAP made the grading result more transparent by showing important words behind the prediction.

Explainable AI can support Khmer short-answer grading, but teachers should remain in control.

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[3] Buoy, R., Chhun, S., Soeurn, C., & Taing, S. (2022). Pretrained models and evaluation data for the Khmer language.

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[12] Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions.

[13] Kumar, V., & Boulanger, D. (2020). Explainable automated essay scoring: deep learning really has pedagogical value.







Thank you for your attention

Q&A

Appendix - LLM hallucination & control

The grader outputs one number between 0 and Max, not free text - so made-up text cannot become a valid score.

You, yesterday | 1 author (You)


```
1  "mode": "bare", "model_version": "google/gemini-3.5-flash-20260519", "eval_date": "2026-06-24", "raw_response": "Forestry: wood products, non-timber forest"}
2  "mode": "bare", "model_version": "google/gemini-3.5-flash-20260519", "eval_date": "2026-06-24", "raw_response": "8"}
3  "mode": "bare", "model_version": "google/gemini-3.5-flash-20260519", "eval_date": "2026-06-24", "raw_response": "5"}
4  "mode": "bare", "model_version": "google/gemini-3.5-flash-20260519", "eval_date": "2026-06-24", "raw_response": "ចំប្រព័ន្ធរដ្ឋបាលនិងច្បាប"}      You, yesterday • no-K
5  "mode": "bare", "model_version": "google/gemini-3.5-flash-20260519", "eval_date": "2026-06-24", "raw_response": "8"}
6  "mode": "bare", "model_version": "google/gemini-3.5-flash-20260519", "eval_date": "2026-06-24", "raw_response": "6"}
7  "mode": "bare", "model_version": "google/gemini-3.5-flash-20260519", "eval_date": "2026-06-24", "raw_response": "ឯទីពារដោះប្រជាជន"}
8  "mode": "bare", "model_version": "google/gemini-3.5-flash-20260519", "eval_date": "2026-06-24", "raw_response": "កម្មគណនាគម\n- កត្តាស"}
```



```
"mode": "reasoning", "model_version": "google/gemini-3.5-flash-20260519", "raw_response": "The student's answer lists the general economic sectors of Cambodia (Industry, Agriculture, Commerce, Tourism, Telecommunications, Weaving) instead of the specific agricultural production systems (Crops, Animal Husbandry, Fishing, Forestry) as required by the question. Therefore, the answer is incorrect.\n\nScore: 0"}
```